

The Core Principles

WHY THIS MODULE MATTERS

Fairness, transparency, accountability, privacy, safety and human oversight are cited everywhere and defined almost nowhere. This module makes each one concrete enough to act on, and shows what it looks like when each is missing.

2.1 Six Principles That Actually Recur

Almost every AI ethics framework in existence — governmental, corporate or academic — converges on a small set of principles. The wording differs; the substance does not. What follows is the version this course uses.

Principles are often dismissed as decoration, and when they are left as single words on a poster that criticism is fair. They become useful only when each is translated into a question someone can actually answer about a specific system. That translation is the purpose of this module.

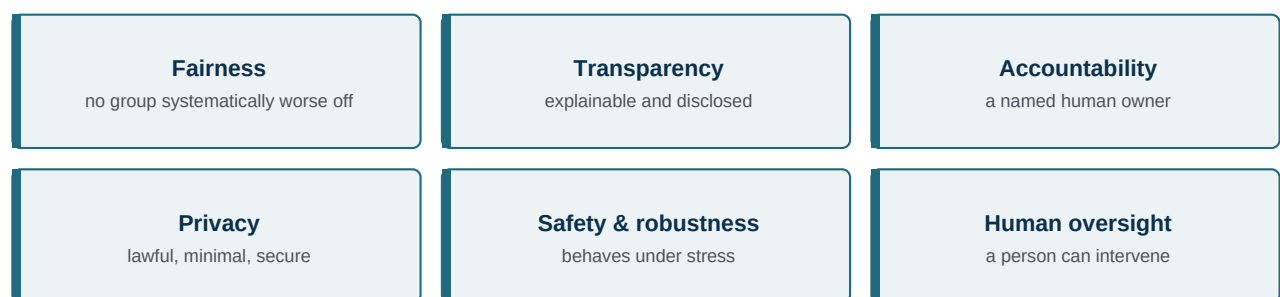


Figure 2.1 — Six core principles referenced throughout this course.

- **Fairness** — the system’s outcomes should not systematically disadvantage people on the basis of protected or proxy characteristics, and any disparity should be identified and either justified or corrected.
- **Transparency** — people affected by a system should be able to know that an automated system is involved and, at an appropriate level, how it works.
- **Accountability** — a named person or function must be answerable for the system’s outcomes. Responsibility must not be diffused until nobody owns it.
- **Privacy** — data used to train and run the system must be collected, stored and used lawfully and proportionately.
- **Safety and reliability** — the system should behave predictably inside its intended scope, and degrade gracefully or fail safely outside it.
- **Human oversight** — a person must be able to review, override or stop the system, and must have the authority, information and time to do so meaningfully.

2.2 Fairness Is a Choice, Not a Property

The most common misconception about fairness is that it is a technical property a system either has or lacks, and that a sufficiently careful engineer can simply verify it. It is not. Fairness is a set of competing definitions, and choosing between them is a judgement about values that cannot be delegated to a model.

Consider a system that predicts which applicants will succeed in a role. You might require that it accepts the same proportion of applicants from each group. Or that it is equally accurate for each group. Or that among those it rejects, the proportion who would actually have succeeded is the same for each group. These are all reasonable

meanings of "fair", and it is mathematically impossible to satisfy all of them at once whenever the underlying groups differ in any respect. Someone has to choose which definition applies, and record why.

The second thing to understand is proxies. Removing a protected characteristic from the training data does not remove its influence, because other variables carry the same information. Postcode carries ethnicity and socioeconomic status. Employment gaps carry parenthood and disability. Membership of certain clubs or societies carries gender. A model with no access to a protected characteristic can reproduce discrimination on it perfectly well, and will do so silently.

CASE STUDY — AMAZON'S ABANDONED RECRUITING TOOL

Between 2014 and 2017 Amazon built an internal tool to score CVs, trained on ten years of applications and hiring outcomes. The intention was straightforward: learn what a successful hire looked like and find more of them.

The historic data reflected a workforce that was overwhelmingly male in technical roles. The model learned accordingly. It penalised CVs containing the word "women's", as in "women's chess club captain", and downgraded graduates of two all-women's colleges. Gender was never an input; the model reconstructed it from proxies.

Amazon's engineers adjusted the model to ignore the specific terms they had found. They then concluded they could not guarantee it had not learned other proxies they had not thought to look for, and the project was abandoned before it was used in live hiring.

The instructive part is not the bias — that was predictable — but the ending. The decision to stop was the governance success. An organisation without the authority to cancel a project at that point would have shipped it.

2.3 Transparency and Explainability

Transparency operates at two levels, and confusing them causes arguments. The first is simply disclosure: telling people that an automated system is involved in a decision affecting them. This is cheap, almost always appropriate, and frequently omitted.

The second is explainability: being able to say why a particular output occurred. This is harder, and the honest position is that for some model types a complete causal explanation is not available. That is not a reason to abandon the principle. It is a reason to match model complexity to the stakes: if a decision materially affects someone's life and you cannot explain it, the correct response is usually to use a simpler model rather than to deploy an unexplainable one and hope.

2.4 Accountability Survives Outsourcing

The single most reliable governance failure is diffusion. The vendor says the model performs to specification. The procurement team says it bought what was asked for. The operational team says it followed the tool's recommendation. The executive says nobody escalated a concern. Every account is true and nobody is accountable.

Accountability means a named individual — not a committee and not a department — who answers for the outcomes of a specific system, and who has the standing to stop it. Buying a system from a third party transfers none of this. If your organisation deploys a decision about a person, your organisation owns that decision, whatever the contract says about the supplier's liability.

2.5 Privacy, Safety and Human Oversight

Privacy is covered in full in Module 5, because in the UK it is largely a legal question rather than an open ethical one. The short version: personal data used to build or run an AI system is subject to the same law as any other personal data, and training a model is not a category exempt from it.

Safety and reliability concern behaviour at the edges. Every model has a scope within which it was validated, and every deployed model eventually meets inputs outside it. The governance question is what happens then — whether the system fails loudly and stops, or fails quietly and continues producing confident nonsense.

Human oversight is the principle most often claimed and least often real. A person who must review two hundred automated recommendations an hour is not providing oversight; they are providing a signature. Meaningful oversight requires that the reviewer has the information needed to disagree, the time to use it, and — critically — that disagreeing is a normal and career-safe act rather than an exception requiring justification.

IN PRACTICE

A useful test for whether human oversight is genuine: look at how often the human overrides the system. If the override rate is essentially zero across thousands of decisions, either the system is flawless or the oversight is nominal. It is almost never the former.

TRY THIS — APPLY IT TO ONE SYSTEM

Take one AI system your organisation uses. For each of the six principles, write a single sentence answering the corresponding question: Who could this disadvantage, and have we checked? Do affected people know it exists? Who is named as accountable? What personal data does it use and on what lawful basis? What does it do with an input it was not built for? Who can override it, and when did they last do so?

Any question you cannot answer in one sentence is the place to start work. Most people find they can answer two or three.

KEY TAKEAWAYS

- Fairness is a choice between mutually incompatible definitions, not a property to be verified — and the choice must be recorded.

KEY TAKEAWAYS

- Removing protected characteristics from training data does not remove bias, because proxy variables carry the same information.
- Transparency has two levels: disclosing that a system is involved, and explaining why it produced a given output. The first is almost always achievable.
- Accountability must sit with a named person who can stop the system, and it is not transferred by buying the system from a vendor.
- Human oversight is only real if the reviewer has time, information and genuine freedom to disagree — a near-zero override rate is a warning sign.

FURTHER READING

- ICO and The Alan Turing Institute — Explaining decisions made with AI
- Equality and Human Rights Commission — guidance on the Equality Act 2010 and automated systems
- Reuters (2018) — reporting on Amazon's abandoned recruiting engine

CHECK-IN QUIZ

1. A recruitment model has no access to age or disability data, but weights unexplained gaps in employment history heavily. The concern this raises is:

- A. Data minimisation
- B. Model drift
- C. Purpose limitation
- D. Proxy discrimination

2. A vendor tells you its model is "certified fair". The most useful follow-up question is:

- A. Which certification body issued that?
- B. Which definition of fairness was used, and what was traded off to achieve it?
- C. What accuracy does the model achieve overall?
- D. How large was the training dataset?

3. An organisation says accountability for its AI systems sits with its Technology Committee. The weakness in this arrangement is that:

- A. Committees meet too infrequently to be useful
- B. Technology functions lack the necessary legal knowledge
- C. Responsibility spread across a committee means no individual answers for outcomes
- D. Committees cannot be held liable in law

4. Which of these is the cheapest and most frequently omitted form of transparency?

- A. Publishing the training dataset
- B. Explaining the model's internal weightings
- C. Providing an appeal to an independent adjudicator

CHECK-IN QUIZ

D. Telling affected people that an automated system is involved

5. A reviewer can override the system, but overriding requires a written justification to a manager while accepting requires one click. The likely effect on oversight is that:

- A. Overrides will be better reasoned and therefore more reliable
- B. The asymmetry discourages disagreement, making oversight nominal
- C. It has no effect, provided the reviewer is trained
- D. It satisfies the Article 22 requirement for human intervention